

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: analytic singular structure plus neural smooth correction

HYBRID GREEN KERNEL

$$G_{\theta} = \text{analytic jump} + \text{analytic cancellation} + \text{learned smooth correction}.$$

ANALYTIC JUMP

Build the Dirac-() structure.

ANALYTIC CANCELLATION

Cancel Heaviside-type contributions.

LEARNED CORRECTION

Represent the remaining smooth residual.

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: analytic singular structure plus neural smooth correction

$$G_{\theta}(t, \eta) = \underbrace{E(t, \eta)M(t)R_{\theta}(t, \eta)}_{\text{learned smooth correction}} + \underbrace{B(t) \left(J_0(t, \eta) - \frac{1}{2}E(t, \eta) \right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t, \eta)}_{\text{Dirac jump}}.$$

DIRAC JUMP

$$G_0(t, \eta) = \begin{cases} t(1 - \eta), & t < \eta, \\ \eta(1 - t), & t \geq \eta, \end{cases} \quad \partial_t^2 G_0 = -\delta(t - \eta).$$

HEAVISIDE CANCELLATION

$$\partial_t J_0(t, \eta) = G_0(t, \eta), \quad S = J_0 - \frac{1}{2}E, \quad \partial_t^2 S = \partial_t G_0.$$

COEFFICIENT FACTORS

$$A(t) = \frac{1}{a_{\text{unit}}(t)}, \quad B(t) = \frac{a'_{\text{unit}}(t) + b_{\parallel, \text{unit}}(t)}{a_{\text{unit}}(t)^2}.$$

GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: analytic singular structure plus neural smooth correction

$$G_{\theta}(t, \eta) = \underbrace{E(t, \eta)M(t)R_{\theta}(t, \eta)}_{\text{learned smooth correction}} + \underbrace{B(t) \left(J_0(t, \eta) - \frac{1}{2}E(t, \eta) \right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t, \eta)}_{\text{Dirac jump}}.$$

LEARNED SMOOTH CORRECTION

$$E(t, \eta)M(t)R_{\theta}(t, \eta), \quad E(t, \eta) = t\eta(1 - \eta), \quad M(t) = 1 - t.$$

GreenNet learns the smooth correction (R_{θ}); the singular and cancellation structure is built into the kernel form.